

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

5 Week: 22 March 2017

Only the Wednesday group has tutorials for this week. Monday and Tuesday groups must do the problems as self-study, and must ask their tutors questions on the prescribed problems at their next meeting.

Slegs die Woensdag groep het tutoriale vir hierdie week. Maandag en Dinsdag groepe moet dié probleme as selfstudie voltooi, en vrae aan hulle tutors rig met die volgende tutoriaal-sessie.

P5.66 (a) $P_{\max} = 48.6 \text{ N}$ and $P_{\min} = 31.7 \text{ N}$ (b) Explanation (c) The block can not slide up the wall. If $P < 62.7 \text{ N}$, it will slide down the wall.

P5.88 Set the static friction coefficient $m_s = \mu_s$.

(a) $M = 3m \sin \theta$ (b) $T_1 = 2mg \sin \theta$, $T_2 = 3mg \sin \theta$ (c) $a = \frac{g \sin \theta}{1+2 \sin \theta}$,

$T_1 = 4mg \sin \theta \left(\frac{1+\sin \theta}{1+2 \sin \theta} \right)$, $T_2 = 6mg \sin \theta \left(\frac{1+\sin \theta}{1+2 \sin \theta} \right)$

(e) $M_{\max} = 3m(\sin \theta + \mu_s \cos \theta)$ (f) $M_{\min} = 3m(\sin \theta - \mu_s \cos \theta)$

(g) $T_{2,\max} - T_{2,\min} = M_{\max}g - M_{\min}g = 6\mu_s mg \cos \theta$

P5.89 (a) Proof (b) If $\tan \theta < \frac{1}{\mu_s}$ is not met, no value of P can move the crate.

P6.3 (a) $8.33 \times 10^{-8} \text{ N}$ inwards (b) $9.15 \times 10^{22} \text{ m/s}^2$ inwards

P6.8 (a) $\vec{T} = (68.6 \text{ N})\hat{i} + (784 \text{ N})\hat{j}$ (b) 0.857 m/s^2

P6.29 3.01 N

P6.45 (a) Explanation (b) FBD (c) 6.05 N (d) 7.82 m/s

P6.50 (a) $v = \sqrt{(6.86 \text{ m/s}^2)R}$ (b) Explanation (c) Explanation

(d) $T = \frac{2\pi R}{\sqrt{Rg \tan 35^\circ}}$ (e) Explanation

P6.59 (a) Proof (b) Explanation (c) Explanation

P6.61 (a) If car is about to slip down the incline, f is directed up the incline and

$v_{\min} = \sqrt{\frac{Rg(\tan \theta - \mu_s)}{1 + \mu_s \tan \theta}}$ If the car is about to slip up the incline, f is directed

down the incline and $v_{\max} = \sqrt{\frac{Rg(\tan \theta + \mu_s)}{1 - \mu_s \tan \theta}}$

(b) $v_{\min} = 0$ and $\mu_s = \tan \theta$